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TP Modul 6

DLL.cpp

```
1  #include <iostream>
2  #include "DLL.H"
3
4  using namespace std;
5
6  void createList_103032400084(List &L) {
7      L.first = nullptr;
8  }
9
10 address allocate_103032400084(intofype x) {
11     address p = new elmList;
12
13     p -> info = x;
14     p -> next = nullptr;
15     p -> prev = nullptr;
16     return p;
17 }
18
19 bool isEmpty_103032400084(List L) {
20     return (L.first == nullptr || L.last == nullptr);
21 }
22
23 void printInfo_103032400084(List L) {
24     address p = L.first;
25     while (p != nullptr) {
26         cout << p -> info;
27         if (p -> next != nullptr) {
28             cout << ", ";
29         }
30         p = p -> next;
31     }
```

```
31
32     cout << endl;
33 }
34
35 void insertFirst_103032400084(List &L, address p){
36     if(isEmpty_103032400084(L)){
37         L.first = p;
38         L.last = p;
39     }else{
40         p -> next = L.first;
41         L.first -> prev = p;
42         L.first = p;
43     }
44 }
45 void insertLast_103032400084(List &L, address p){
46     if(isEmpty_103032400084(L)){
47         L.first = p;
48         L.last = p;
49     }else{
50         p -> prev = L.last;
51         L.last -> next = p;
52         L.last = p;
53     }
54 }
55 void deleteFirst_103032400084(List &L, address &p){
56     if(isEmpty_103032400084(L)){
57         p = nullptr;
58     }else if(L.first == L.last){
59         p = L.first;
60         L.first = nullptr;
```

```

61         L.last = nullptr;
62     }else{
63         p = L.first;
64         L.first = L.first->next;
65         L.first->prev = nullptr;
66         p->next = nullptr;
67     }
68 }
69 void deleteLast_103032400084(List &L, address &p){
70     if(isEmpty_103032400084(L)){
71         p = nullptr;
72     }else if(L.first == L.last){
73         p = L.first;
74         L.first = nullptr;
75         L.last = nullptr;
76     }else{
77         p = L.last;
78         L.last = L.last->prev;
79         L.last->next = nullptr;
80         p->prev = nullptr;
81     }
82 }
83 void printInfoFrontToBack_103032400084(List L){
84     address p = L.first;
85     while (p != nullptr) {
86         cout << p -> info;
87         if (p -> next != nullptr) {
88             cout << ", ";
89         }
90         p = p -> next;

```

```

91     }
92
93     cout << endl;
94 }
95 void printInfoBackToFront_103032400084(List L){
96     address p = L.last;
97     while (p != nullptr) {
98         cout << p -> info;
99         if (p -> prev != nullptr) {
100             cout << ", ";
101         }
102         p = p -> prev;
103     }
104
105     cout << endl;
106 }
107

```

DLL.H

```
1  #ifndef DLL_H_INCLUDED
2  #define DLL_H_INCLUDED
3
4  #include <iostream>
5
6  using namespace std;
7
8  typedef int infotype;
9  typedef struct elmlist *address;
10
11  struct elmlist {
12      infotype info;
13      address next;
14      address prev;
15  };
16
17  struct List {
18      address first;
19      address last;
20  };
21
22  void createList_103032400084(List &L);
23  bool isEmpty_103032400084(List L);
24  address allocate_103032400084(infotype x);
25  void printInfo_103032400084(List L);
26  void insertFirst_103032400084(List &L, address p);
27  void insertLast_103032400084(List &L, address p);
28  void deleteFirst_103032400084(List &L, address &p);
29  void deleteLast_103032400084(List &L, address &p);
30  void printInfoFrontToBack_103032400084(List L);
31  void printInfoBackToFront_103032400084(List L);
32  void insertLast2(List &L, address p);
33
34  #endif // DLL_H_INCLUDED
35
```

main.cpp (2.2)

```
1  #include <iostream>
2  #include "DLL.H"
3
4  using namespace std;
5
6  int main() {
7      List L;
8      int angka1, angka2, angka3;
9      address ak1, ak2, ak3;
10
11     createList_103032400084(L);
12
13     cout << "Masukkan elemen pertama: ";
14     cin >> angka1;
15
16     ak1 = allocate_103032400084(angka1);
17     insertFirst_103032400084(L, ak1);
18
19     cout << "Masukkan elemen kedua di awal: ";
20     cin >> angka2;
21
22     ak2 = allocate_103032400084(angka2);
23     insertFirst_103032400084(L, ak2);
24
25     cout << "Masukkan elemen ketiga di akhir: ";
26     cin >> angka3;
27
28     ak3 = allocate_103032400084(angka3);
29     insertLast_103032400084(L, ak3);
30
31     cout << "daftar elemen list: ";
32     printInfo_103032400084(L);
33
34     return 0;
35 }
36
```

Input & output (2.2)

```
Masukkan elemen pertama: 10
Masukkan elemen kedua di awal: 5
Masukkan elemen ketiga di akhir: 20
daftar elemen list: 5, 10, 20
```

main.cpp (3.2)

```
1  #include <iostream>
2  #include "DLL.H"
3
4  using namespace std;
5
6  int main() {
7      List L;
8      int angka1, angka2, angka3;
9      address ak1, ak2, ak3;
10
11      createList_103032400084(L);
12
13      cout << "Masukkan elemen pertama: ";
14      cin >> angka1;
15
16      ak1 = allocate_103032400084(angka1);
17      insertFirst_103032400084(L, ak1);
18
19      cout << "Masukkan elemen kedua di awal: ";
20      cin >> angka2;
21
22      ak2 = allocate_103032400084(angka2);
23      insertLast_103032400084(L, ak2);
24
25      cout << "Masukkan elemen ketiga di akhir: ";
26      cin >> angka3;
27
28      ak3 = allocate_103032400084(angka3);
29      insertLast_103032400084(L, ak3);
30      cout << endl;
31
32      deleteFirst_103032400084(L, ak1);
33      cout << "elemen pertama telah dihapus" << endl;
34      deleteLast_103032400084(L, ak3);
35      cout << "elemen terakhir telah dihapus" << endl << endl;
36
37      cout << "daftar elemen list: ";
38      printInfo_103032400084(L);
39
40      return 0;
41  }
42
```

Input & output (3.2)

```
Masukkan elemen pertama: 10
Masukkan elemen kedua di akhir: 15
Masukkan elemen ketiga di akhir: 20

elemen pertama telah dihapus
elemen terakhir telah dihapus

daftar elemen list: 15
```

main.cpp (4.2)

```
1  #include <iostream>
2  #include "DLL.H"
3
4  using namespace std;
5
6  int main() {
7      List L;
8      int angka1, angka2, angka3, angka4;
9      address ak1, ak2, ak3, ak4;
10
11     createList_103032400084(L);
12
13     cout << "Masukkan elemen di akhir: ";
14     cin >> angka1;
15
16     ak1 = allocate_103032400084(angka1);
17     insertLast_103032400084(L, ak1);
18
19     cout << "Masukkan elemen di akhir: ";
20     cin >> angka2;
21
22     ak2 = allocate_103032400084(angka2);
23     insertLast_103032400084(L, ak2);
24
25     cout << "Masukkan elemen di akhir: ";
26     cin >> angka3;
27
28     ak3 = allocate_103032400084(angka3);
29     insertLast_103032400084(L, ak3);
30
```

```

31     cout << "Masukkan elemen di akhir: ";
32     cin >> angka4;
33
34     ak4 = allocate_103032400084(angka4);
35     insertLast_103032400084(L, ak4);
36     cout << endl;
37
38     cout << "daftar elemen dari depan ke belakang: ";
39     printInfoFrontToBack_103032400084(L);
40     cout << endl;
41
42     cout << "daftar elemen dari belakang ke depan: ";
43     printInfoBackToFront_103032400084(L);
44
45
46     return 0;
47 }
48

```

Input & output (4.2)

```

Masukkan elemen di akhir: 1
Masukkan elemen di akhir: 2
Masukkan elemen di akhir: 3
Masukkan elemen di akhir: 4

daftar elemen dari depan ke belakang: 1, 2, 3, 4
daftar elemen dari belakang ke depan: 4, 3, 2, 1

```